

Haixi Zhao

Email: zhaohaixi2005@gmail.com | Phone: +86 199 7666 0738

[Research Website](#)

EDUCATION BACKGROUND

Huazhong University of Science and Technology (HUST)

Sept. 2023 - Jun. 2027(Expected)

Sino-French Joint Dual-Degree Program with **Université Paris-Saclay, Paris, France**

- Bachelor of Science in **Bioscience**
- **Cumulative GPA: 4.06/5.0(CN) ; 17.449/20(FR)**
- Related Coursework: Mathematics for Biology I(98) Deep learning(98) Mathematics for Biology II(96) Genetics(96) BioStatistics(96) Chemistry-Biology(90) PPEI(100) Most Laboratory Course(90+)
- Honors & Awards: Scholarship for Scientific and Technological Innovation (2025&2026) Award for Self-Reliance and Excellence (2024&2025&2026) Community Service Scholarship (2025&2026) Outstanding Leader, Spark Program <Undergraduate Track> (2024) BAIDU Technology Certification-LLM Prompt Engineer(2025) Modeling Leader of HUST-China, iGEM Gold Medal & Best iGemers (2025) Team Leader, 2nd Prize – 10th China Undergraduate Life Sciences Competition (2024)

PUBLICATIONS

[1] **Haixi Zhao**, Jiarong Li, Yilin Huang, Nan Xiang, Chen Jin, Ruijian Shao, and Su Sun. (2026). From oxygen detoxification to peroxide supply: Microbial NADH oxidases in redox physiology and H₂O₂-coupled biocatalysis. *Microbiological Research*, 128741. <https://doi.org/10.1016/j.micres.2026.128741> (SCI, IF = 8.5, JCR Q1, CAS Q1 Top Journal)

[2] Bo Wei, **Haixi Zhao**, Shi Chang, and Wenlong Wang (2026). The NAT10/c-Myc Positive Feedback Loop Orchestrates tRNA ac⁴C Modification and OTUB1-Mediated Protein Stabilization to Drive Anaplastic Thyroid Carcinoma Progression. *Cellular & Molecular Biology Letters*, <https://doi.org/10.1186/s11658-026-00977-x> (SCI, IF = 10.2, JCR Q1, CAS Q1 Top Journal)

[3] Wenlong Wang*, Ying Ding*, **Haixi Zhao**, Shouman Wang, Juan Huang, and Lunquan Sun (2025). NSUN2-tRNA Val-CAC-axis-regulated codon-biased translation drives triple-negative breast cancer glycolysis and progression. *Cellular & Molecular Biology Letters*, 30:100, <https://doi.org/10.1186/s11658-025-00781-z> (SCI, IF = 10.2, JCR Q1, CAS Q1 Top Journal)

Manuscripts under review

[4] Ruijian Shao, Zhenxiong Zhang, Liduo Lu, Yixue Zhou, Yilin Huang, **Haixi Zhao**, Fuying Ma, Shangxian Xie, and Hongbo Yu. An enzyme cascade of pre-oxidation and in situ H₂O₂ generation enables LiP-driven lignin dimer depolymerization. Submitted to *Journal of Agricultural and Food Chemistry*.

RESEARCH EXPERIENCE

Development of RFD3-Mosaic: A Programmable Generative Framework

for Protein Assembly

[Khmelinskaia Group | Member](#)

LMU Munich, Germany Jul. 2026 – Present Research Intern — Prof. Alena Khmelinskaia's Lab
Motivation: Current protein backbone generators support motif- or symmetry-conditioned generation but lack a unified representation of oligomeric architecture spanning structural components, connectivity, interaction geometry, and global topology. Developed RFD3-Mosaic to formulate protein assembly design as a programmable constrained-generation problem.

- Conceived and developed RFD3-Mosaic, which compiles declarative specifications of components, motifs, connectivity, interaction constraints, symmetry, and topology into executable RFD3 conditioning programs.
- Reformulated backbone diffusion over symmetry-coupled residue orbits and constraint groups, propagating motifs, interfaces, and higher-order geometric relationships consistently throughout sampling.
- Introduced group-constrained SE(3) mobility and local-neighborhood denoising, allowing fixed, mobile, or partially constrained elements to adapt to emerging scaffolds while preserving internal geometry, inter-component relationships, and target symmetry.
- Generalized the framework across de novo oligomer generation, motif- and interface-conditioned scaffolding, multi-interface design, and cyclic, dihedral, polyhedral, and helical symmetries, providing a unified route to rings, cages, and higher-order protein assemblies.

De Novo Design of PTM Capture Binders

[Chu Wang Lab | Research](#)

Peking University Beijing, China Dec. 2025 – Present Research Intern — Prof. Chu Wang's Lab
Motivation: Aimed to identify the spatial occurrence and modification sites of specific PTMs, with a focus on lysine-linked LCA (Lys-LCA) modifications, by developing de novo designed capture binders to selectively enrich PTM-modified species, enabling precise localization and characterization of modification states.

- Designed all-helical pseudocyclic binders for PTM capture, enabling structurally constrained recognition through shape complementarity and topological closure.
- Built a generative design pipeline integrating BoltzGen, RFDiffusion, and LigandMPNN for backbone generation, topology control, and sequence optimization.
- Implemented geometry-aware filtering for structural stability, helix packing quality, and binding pocket formation.
- Established an in silico screening and ranking workflow to prioritize candidates for downstream experimental evaluation.

De Novo Design of Functional Membrane Proteins

Westlake University Hangzhou, China Jun. 2025 -Sept.2025

Peilong Lu Laboratory | Alumni

Research Intern — Prof. Peilong Lu's Lab

Motivation: Aimed to resolve the trade-off between flexibility and stability in de novo membrane protein design by engineering bridge architectures that couple distinct conformational states with multifunctional activity, enabling controllable gating and programmable protein function.

- Benchmarked AlphaFold2.3, AlphaFold3, Boltz-2, and Chai-1 across template-free and template-guided settings, using template-guided AF3 to recover distinct open and closed conformations from collapsed de novo predictions and generate state-specific structural seeds for subsequent bridge design.
- Applied RFdiffusion to design transmembrane helices, soluble domains, and bridge architectures that connect and stabilize alternative conformational states, while tuning guiding potentials and filtering candidates for global stability and lipid compatibility.
- Performed molecular cloning, protein expression and purification, and cryo-EM sample preparation toward experimental validation of the folding accuracy and structural stability of selected designs.

Structural Optimization of NOX Enzymes for Enhanced H₂O₂ Production via Directed Evolution

Huazhong University of Science and Technology

Wuhan, China

Mar. 2024 – Present

Motivation: Aimed to address the challenge of enhancing NADH oxidase catalytic efficiency to achieve sustainable NADH regeneration and controlled H₂O₂ production, enabling efficient coupling with lignin-degradation systems.

- Screened NOX isoforms and enzymatic reaction systems to establish efficient NAD⁺ regeneration and H₂O₂ co-production for lignin degradation.
- Integrated MEGA-X-based sequence alignment, molecular docking, thermodynamic binding-affinity modeling, and validation to identify residues governing substrate specificity and NADH/FAD recognition.
- Applied molecular dynamics, saturation mutagenesis, and directed evolution to remodel NOX–FAD interfaces, strengthen cofactor retention, and prioritize variants with enhanced catalytic efficiency and H₂O₂ yield.

Profiling Hypoxia Responses in Colonic Organoids Using Microfluidic Platforms

Huazhong University of Science and Technology

Wuhan, China

Oct. 2024 - Mar. 2025

Motivation: Sought to investigate how hypoxia and nutrient deprivation reshape intestinal organoid cell states by using microfluidic platforms to recapitulate pathological microenvironments.

- Optimized scRNA-seq preprocessing and quality-control parameters, including mitochondrial-content threshold tuning, to retain stress-responsive cells and improve detection of hypoxia-associated transcriptional signatures.
- Refined PCA/UMAP dimensionality-reduction and marker-based annotation workflows to improve cluster resolution and resolve distinct cell subpopulations.
- Analyzed differentially expressed genes and stress-response programs to characterize cell-state remodeling induced by hypoxia and nutrient deprivation in microfluidic organoid models.

tRNA Modification-Mediated Translational Regulation in Cancer

Xiangya Hospital, Central South University

Changsha, China

Jun. 2024 - Feb. 2026

Motivation: Investigated how NSUN2- and NAT10-mediated tRNA modifications regulate codon-biased translation, oncogenic protein output, and protein stability during cancer progression.

- Analyzed codon-usage and translation-efficiency data to characterize how the NSUN2–tRNA Val-CAC and NAT10–tRNA ac⁴C axes reshape oncogenic protein output in TNBC and ATC.
- Built structure-based models of NSUN2–tRNA Val-CAC and OTUB1–c-Myc interactions using AlphaFold3 prediction, molecular docking, and interface analysis.
- Integrated structural and molecular evidence to support a mechanistic model linking NAT10-mediated tRNA ac⁴C regulation, OTUB1-dependent c-Myc stabilization, and ATC progression.

COMPETITION

iGEM 2025 — HUST-China (Modeling Leader)

Modeling page: <https://2025.igem.wiki/hust-china/model>

Gold Medal & Best iGemers (2025)

Sept.2024 – Nov.2025

- Built the TriScale digital-twin modeling workflow, integrating molecular docking, AI-guided enzyme redesign, and growth-curve machine learning for system-level pathway design.
- Predicted PEX5–PTS1 binding affinities to guide signal-peptide selection and reduce uncertainty in peroxisomal-import efficiency.
- Redesigned the rate-limiting enzyme t-HMGR to improve predicted catalytic stability and modeled *Y. lipolytica* growth using Gompertz + RF/XGBoost to optimize culture conditions.

10th China Undergraduate Life Sciences Competition -- CULSC (Team Leader)

Apr.2024 – Jun.2025

- Led the team to complete a directed-evolution project based on the NOX system, coordinating experimental design, data analysis, and project integration under competition constraints.
- Addressed key workflow bottlenecks—including construct efficiency and experimental reproducibility—and guided the team through final documentation and presentation to achieve the **2nd Prize**.

TECHNICAL SKILLS

- **Languages:** English (IELTS 6.5) ,Chinese (native)
- **Programming Languages:** Python, C++, R, Linux, HPC (Slurm)
- **Software & Tools:**
 - Structural Modeling & Design:** AlphaFold2/3 , Boltz2/gen , Chai-1 , RFdiffusion , ProteinMPNN , LigandMPNN , Rosetta , PyRosetta , Protenix
 - Molecular editor & visualization:** PyMOL , ChimeraX , Open Babel , RELION
 - Docking & Interaction Modeling:** AutoDock4 & Vina , RosettaDock , HADDOCK , ClusPro
 - Genomics & Bioinformatics:** Scanpy , SnapGene
 - Molecular dynamics:** Gromacs , VMD
- **Wet lab skills:**
 - Molecular Biology:** PCR, multiplex PCR, qPCR,
 - Microbiology:** Bacterial culture, plasmid propagation, transformation
 - Cryo-EM:** Grid preparation, vitrification, data collection
 - Protein Biochemistry & Structural Biology:** SDS-PAGE, Western blot, Ni-NTA/His-tag, SEC, ÄKTA FPLC